Analog–Digital Conversion Techniques

Summarized Notes: PCM, Delta Modulation, ASK, FSK, PSK & QAM

1. Analog-to-Digital Conversion

1.1 Pulse Code Modulation (PCM)

Pulse Code Modulation is the most common technique used to convert an analog signal into digital data. It works by capturing an analog waveform through three sequential processes — sampling, quantizing, and encoding — so that the continuous signal can be represented as a stream of binary digits suitable for digital transmission and storage.

Advantages

- Highly resistant to noise and signal distortion since it deals with discrete digital values.
- Allows easy multiplexing, regeneration, and error detection/correction.
- Produces good voice/audio quality when enough bits per sample are used.
- Signal can be regenerated at repeaters without accumulating noise.

Disadvantages

- Requires a high bandwidth compared to the original analog signal.
- Complex encoder/decoder circuitry compared to simpler modulation schemes like DM.
- Quantization introduces an inherent error (quantization noise) that cannot be fully eliminated.

1.2 Components of a PCM Encoder

A PCM encoder converts the analog input into the PAM (Pulse Amplitude Modulated) signal via sampling, then quantizes the sampled amplitudes into a fixed number of levels, and finally encodes each quantized value into an nb-bit binary code word, producing the final digital data stream.

1.3 Sampling

Sampling is the first step of PCM, where the analog signal is measured at regular intervals Ts (the sample interval or period). The inverse of this interval is called the sampling rate or sampling frequency, fs = 1/Ts. There are three sampling methods: ideal sampling, natural sampling, and flat-top sampling.

$$fs = 1 / Ts$$

- **Ideal sampling:** Instantaneous pulses are taken from the analog signal; theoretically ideal but practically impossible to implement.
- **Natural sampling:** A high-speed switch is turned on only for the small period during which sampling occurs, producing a sequence of samples that retains the shape of the analog signal.
- **Flat-top sampling (Sample and Hold):** The most commonly used method; a circuit creates flat-top samples. This process is also referred to as Pulse Amplitude Modulation (PAM).

1.4 Sampling Rate — Nyquist Theorem

According to the Nyquist theorem, the sampling rate must be at least twice the highest frequency contained in the analog signal to allow accurate reconstruction. A signal can only be sampled if it is band-limited. For a low-pass signal, the bandwidth equals the highest frequency, while for a bandpass signal, the bandwidth is lower than the maximum frequency value.

$$Nyquist\ Rate = 2 \times f_{max}$$

- **Nyquist rate sampling** (fs = 2f): the sine wave can be exactly recovered.
- **Oversampling** (fs = 4f): more samples than required, wave is recovered accurately.
- **Undersampling** (fs = f): insufficient samples, leading to distortion/aliasing and loss of original signal shape.

1.5 Quantization

Quantization is the process of mapping the infinite range of sampled amplitude values into a finite set of discrete levels, since infinite/nonintegral amplitude values cannot be used directly in encoding. The amplitude range between Vmin and Vmax is divided into L zones, each of height Δ (delta), and every sample amplitude is approximated to the quantized value at the midpoint of its zone.

$$\Delta = (Vmax - Vmin) / L$$

- Quantized values are assigned integers from 0 to L–1, corresponding to the midpoint of each zone.
- **Quantization error** is the difference between the actual sample value and its quantized value; it is an inherent approximation error in PCM.
- The Signal-to-Noise Ratio due to quantization depends on the number of bits per sample (nb): SNRdB = 6.02nb + 1.76 dB.
- **Types of quantization:** Uniform Quantization (equal-sized zones) and Non-Uniform Quantization (unequal zone sizes, often used to reduce error for smaller-amplitude signals).

1.6 Encoding

After a sample is quantized, it is converted into an nb-bit binary code word (e.g., quantization level 2 becomes 010, level 5 becomes 101). The number of bits per sample is determined by the number of quantization levels L, using nb = log2(L). The overall bit rate of the PCM output is the product of the sampling rate and the number of bits per sample.

$$Bit\ Rate = Sampling\ Rate\ (fs) \times Bits\ per\ Sample\ (nb)$$

1.7 Original Signal Recovery (PCM Decoder)

The PCM decoder reconstructs the original analog signal from the digital data. It first converts the incoming code words into pulses that hold their amplitude until the next pulse, forming a staircase-shaped signal. This staircase signal is then passed through a low-pass filter — with the same cutoff frequency as the original signal — to smooth it back into a continuous analog waveform. If the signal was sampled at or above the Nyquist rate with sufficient quantization levels, the original signal can be accurately recreated.

1.8 Delta Modulation (DM)

Delta Modulation is a simplified alternative to PCM. While PCM encodes the absolute amplitude value of each sample, DM only encodes the change (delta, δ) between the current sample and the previous one. If the change is positive, a 1 is recorded; if negative, a 0 is recorded, producing a staircase-approximation of the original analog signal.

Advantages

- Much simpler to implement than PCM, requiring only 1 bit per sample.
- Lower bit rate/bandwidth requirement compared to PCM for equivalent sampling.
- Simple encoder and decoder circuitry (comparator, staircase maker, delay unit).

Disadvantages

- Prone to slope overload distortion when the analog signal changes faster than the staircase can follow.
- Prone to granular noise when the signal is relatively constant but the staircase keeps oscillating by δ.
- Generally lower accuracy/quality than PCM unless adaptive techniques are used.

1.9 DM Modulator

The DM modulator, used at the sender, compares the analog input signal with a gradually built staircase signal at each sampling interval. If the analog signal's amplitude is larger than the staircase value, the next output bit is 1 (and the staircase maker moves the staircase up by δ); otherwise, the bit is 0 (and the staircase moves down by δ). A delay unit holds the staircase value between two comparisons.

1.10 DM Demodulator

The DM demodulator reconstructs the analog signal from the incoming digital bit stream using a staircase maker and a delay unit, similar in structure to the modulator. The resulting staircase signal is then passed through a low-pass filter to smooth it into a continuous analog output signal.

1.11 Adaptive Delta Modulation (ADM)

In Adaptive Delta Modulation, the step size δ is not fixed but varies dynamically according to the amplitude and rate of change of the analog signal. This adaptive adjustment significantly reduces both slope overload distortion and granular noise, resulting in a much smaller quantization error compared to standard DM (and even standard PCM in some cases).

2. Digital-to-Analog Conversion

2.1 Introduction to Digital-to-Analog Conversion

Digital-to-analog conversion is the process of changing one of the characteristics (amplitude, frequency, or phase) of an analog carrier signal based on the information contained in digital data, so that digital data can be transmitted over analog communication links.

- Three basic mechanisms: Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK).
- A combined mechanism that varies both amplitude and phase together is called Quadrature Amplitude Modulation (QAM), which is the most efficient and most commonly used technique today.
- **Bit rate** is the number of bits transmitted per second, while **Baud rate** is the number of signal elements transmitted per second. In analog transmission of digital data, baud rate $\leq$ bit rate.

- Relationship: S = N × (1/r), where S = baud rate, N = bit rate, and r = number of data bits carried per signal element.

2.2 Amplitude Shift Keying (ASK)

In Amplitude Shift Keying, the amplitude of the carrier signal is varied to represent digital data while the frequency and phase remain constant. ASK is normally implemented using only two amplitude levels, known as Binary ASK or On-Off Keying (OOK), where the peak amplitude of one signal element is 0 and the other equals the amplitude of the carrier frequency. It is implemented by multiplying the unipolar NRZ digital signal by the carrier signal from an oscillator.

Bandwidth: $B = (1 + d) \times S$

Advantages

- Simple to implement and demodulate.
- Requires less bandwidth than FSK for the same bit rate.
- Low-cost transmitter and receiver design.

Disadvantages

- Highly susceptible to noise and amplitude-based interference, since noise easily affects amplitude.
- Lower efficiency and poorer performance in noisy channels compared to FSK and PSK.
- Not favoured for high-speed or long-distance data transmission.

2.3 Frequency Shift Keying (FSK)

In Frequency Shift Keying, the frequency of the carrier signal is varied to represent digital data, while the peak amplitude and phase remain constant. The modulated signal's frequency stays constant during one signal element but changes when the data element changes.

2.4 Binary FSK (BFSK)

Binary FSK uses two carrier frequencies, f1 and f2 — one carrier represents data bit 0, and the other represents data bit 1. These carrier frequencies are very high with only a very small difference between them. The middle of one bandwidth is f1 and the other f2, both Δf apart from the midpoint between the two bands, giving a frequency difference of 2Δf.

Bandwidth: $B = (1 + d) \times S + 2\Delta f$

Advantages

- Less susceptible to noise than ASK since frequency is harder for noise to distort than amplitude.
- Better noise immunity makes it suitable for radio-frequency (wireless) transmission.
- Simple detection using frequency discriminators/filters.

Disadvantages

- Requires more bandwidth than ASK due to the two separate carrier frequency bands.
- More complex circuitry (two oscillators or a VCO) than ASK.
- Noncoherent BFSK can suffer from phase discontinuity between signal elements.

2.5 Implementation of BFSK

There are two implementation types of BFSK: noncoherent and coherent. In noncoherent BFSK, there may be a discontinuity in phase where one signal element ends and the next begins; it can be implemented by treating BFSK as two separate ASK modulations using two different carrier frequencies. In coherent BFSK, the phase continues smoothly through the boundary of two signal elements, implemented using a single Voltage-Controlled Oscillator (VCO) whose frequency changes according to the input voltage — remaining at

a regular frequency for amplitude 0, and increasing for a positive amplitude.

2.6 Multilevel FSK (MFSK)

Multilevel FSK uses more than two carrier frequencies to send multiple bits per signal element — for example, four frequencies (f1–f4) send 2 bits at a time, while eight frequencies send 3 bits at a time. This increases the number of bits transmitted per signal element, improving bandwidth efficiency at the cost of a more complex demodulator.

Bandwidth: $B = (1 + d) \times S + (L - 1)2\Delta f \rightarrow B = L \times S$

2.7 Phase Shift Keying (PSK)

In Phase Shift Keying, the phase of the carrier signal is varied to represent two or more different signal elements, while the peak amplitude and frequency remain constant. PSK is more common today than ASK or FSK, and QAM (which combines ASK and PSK) is the dominant digital-to-analog modulation method used in practice.

Advantages

- Less susceptible to noise than ASK, since noise affects amplitude more easily than phase.
- Superior to FSK because it does not require two separate carrier signals — saving bandwidth.
- Higher bandwidth efficiency, allowing more bits per signal element (e.g., in QPSK).

Disadvantages

- Requires more sophisticated hardware/circuitry to accurately distinguish between different phases.
- Performance is limited by the equipment's ability to detect small phase differences, restricting achievable bit rate.
- More complex synchronization requirements at the receiver compared to ASK.

2.8 Binary PSK (BPSK)

Binary PSK is the simplest form of PSK, using only two signal elements: one with a phase of 0° and the other with a phase of 180°. It is implemented by multiplying a polar NRZ signal with the carrier frequency — bit 1 (positive voltage) produces a phase starting at 0°, and bit 0 (negative voltage) produces a phase starting at 180°, since the 180° element is simply the complement of the 0° element.

Bandwidth: $B = (1 + d) \times S$

2.9 Quadrature PSK (QPSK)

QPSK uses two separate BPSK modulations combined together — one in-phase and one quadrature (out-of-phase). Incoming bits are first split via a serial-to-parallel converter, sending alternating bits to each of the two modulators; if the original bit duration is T, each BPSK signal receives bits of duration 2T (half the frequency). The two resulting sine waves are combined to produce an output signal with one of four possible phases: 45°, –45°, 135°, and –135°, giving four signal element types (L = 4) and 2 bits per signal element (r = 2).

2.10 Constellation Diagram

A constellation diagram is a graphical tool used to represent the amplitude and phase of a signal element when two carriers (in-phase and quadrature) are used together. Each possible signal element type is plotted as a dot, where the distance of the dot from the origin represents amplitude and the angle represents phase. For example, ASK (OOK) has two points along the axis, BPSK has two points 180° apart, and QPSK has four points evenly spaced around a circle representing the bit pairs 00, 01, 10, 11.

2.11 Quadrature Amplitude Modulation (QAM)

Quadrature Amplitude Modulation combines both ASK and PSK by using two carriers — one in-phase and one quadrature — each with different, independently varied amplitude levels. This overcomes the limitation of PSK, where equipment can only distinguish a limited number of phase differences, allowing QAM to pack more bits into each signal element (e.g., 4-QAM, 16-QAM). The minimum bandwidth required for QAM transmission is the same as that required for ASK and PSK transmission.

Advantages

- Highest bandwidth efficiency among ASK, FSK, and PSK — carries more bits per signal element.

- Combines the strengths of both amplitude and phase modulation for higher data rates.
- Widely used in modern high-speed communication systems (e.g., modems, Wi-Fi, digital cable).

Disadvantages

- More susceptible to noise than PSK alone, since it also relies on amplitude variations.
- Requires more complex and precise transmitter/receiver hardware to distinguish many amplitude-phase combinations.
- Higher signal-to-noise ratio (SNR) required to reliably detect closely spaced constellation points.

2.12 Quick Comparison: ASK vs FSK vs PSK

Parameter	ASK	FSK	PSK
Varies	Amplitude	Frequency	Phase
Constant	Freq. & Phase	Amplitude & Phase	Amplitude & Freq.
Noise Immunity	Low	Moderate	High
Bandwidth	B=(1+d)S	B=(1+d)S+2Δf	B=(1+d)S
Complexity	Low	Moderate	High